INCH-POUND
MIL-DTL-32687/2
w/Amendment 3
29 May 2026
SUPERSEDING
MIL-DTL-32687/2
w/Amendment 2
18 July 2025

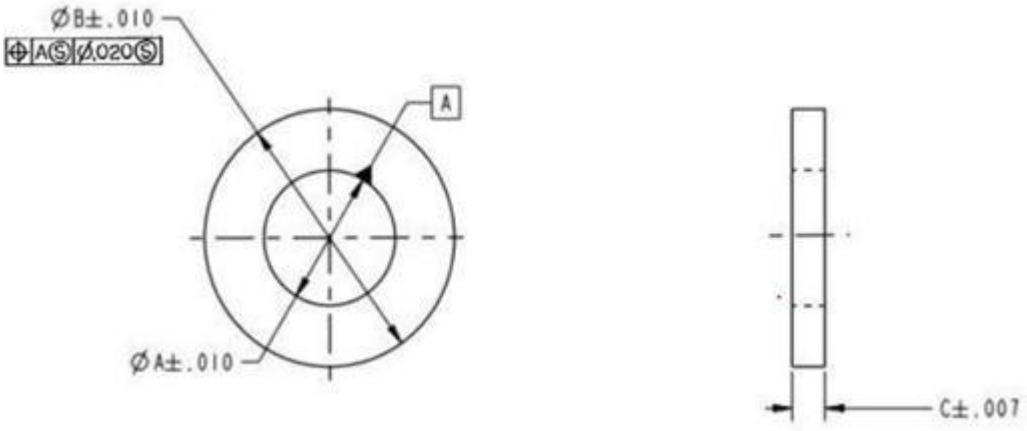
# DETAIL SPECIFICATION SHEET

# WASHER, FLAT, ALLOY OR CARBON STEEL, ROUND, HARDENED, ZINC-NICKEL PLATED, EXTRA LARGE

## (Similar to 19207-10941915)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32687.




NOTES:
1. Dimensions are in inches.
2. If stamping is used to manufacture washers, tolerance for dimensions A and B shall be + 0.040/ -0.010.
3. All edges and corners shall be free from burrs.
4. Unless otherwise specified, edges shall be broken R 0.02+ 0.02/ - 0.00.

FIGURE 1. Washer, flat, alloy or carbon steel, hardened, extra large.

Comments, recommendations, additions, deletions, clarifications and any data that may improve this document should be sent to U.S. Army DEVCOM Ground Vehicle System Center, ATTN: FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Detroit Arsenal, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil/.

AMSC N/A FSC 5310

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:34Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

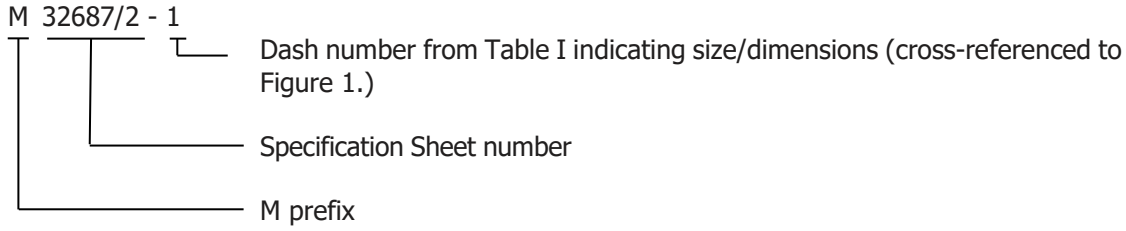
1 Steel composition. Unless otherwise specified, washers shall be made from one of the following materials:

Steel, alloy, strip, sheet or plate, grade or designation 4130, 4140, or 8640, spec ASTM A506, or ASTM A829/A829M
Steel, carbon, strip, designation 1050, spec ASTM A684/A684M
Steel, alloy, bar, grade 4130, 4140, 8640, or 8740, spec ASTM A108 or ASTM A322
Steel, alloy, strip, sheet or plate, 4130, spec SAE AMS6350, SAE AMS6351 or SAE AMS6345

2 Hardness. Washers shall be quenched and tempered to Rockwell C33 through 38 (or equivalent hardness scale).

3 Finish. Zinc-nickel plating per MIL-PRF-32647, Type K20, or zinc-nickel plating per ASTM F1941/F1941M, Fe/Zn-Ni 8BNS.

4 Part number. Part or Identifying Number (PIN) consists of the following:




Example: M32687/2-1 is a Flat Washer – Nominal Size 0.250, ID 0.281, OD 0.750, and 0.090 inch thickness with zinc-nickel, Type K20 Plating per MIL-PRF-32647, or Zinc-Nickel per ASTM F1941/F1941M, Fe/Zn-Ni 8BNS. (See Table II for cross-reference from part number 19207-10941915 to PIN).

TABLE I. Dimensions.

| PIN | Nominal Size | A | B | C |
|---|---|---|---|---|
| M32687/2-1 | 0.250 | 0.281 | 0.750 | 0.090 |
| M32687/2-2 | 0.313 | 0.343 | 0.875 | 0.090 |
| M32687/2-3 | 0.375 | 0.406 | 1.000 | 0.104 |
| M32687/2-4 | 0.438 | 0.468 | 1.250 | 0.119 |
| M32687/2-5 | 0.500 | 0.531 | 1.375 | 0.134 |
| M32687/2-6 | 0.563 | 0.594 | 1.500 | 0.164 |
| M32687/2-7 | 0.625 | 0.656 | 1.750 | 0.179 |
| M32687/2-8 | 0.750 | 0.812 | 2.000 | 0.194 |
| M32687/2-9 | 0.875 | 0.938 | 2.250 | 0.250 |
| M32687/2-10 | 1.000 | 1.062 | 2.500 | 0.280 |
| M32687/2-11 | 1.125 | 1.250 | 2.750 | 0.320 |
| M32687/2-12 | 1.250 | 1.375 | 3.000 | 0.360 |
| M32687/2-13 | 1.375 | 1.500 | 3.250 | 0.400 |
| M32687/2-14 | 1.500 | 1.625 | 3.500 | 0.450 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:34Z
Check the source to verify that this is the current version before use.

TABLE II. Reference and Zn-Ni new part numbers. /1, /3

| Reference Part Number (Zn-HC) /2, /4 | New Part Number (Zn-Ni) /4 |
|---|---|
| 10941915-1 | M32687/2-1 |
| 10941915-2 | M32687/2-2 |
| 10941915-3 | M32687/2-3 |
| 10941915-4 | M32687/2-4 |
| 10941915-5 | M32687/2-5 |
| 10941915-6 | M32687/2-6 |
| 10941915-7 | M32687/2-7 |
| 10941915-8 | M32687/2-8 |
| 10941915-9 | M32687/2-9 |
| 10941915-10 | M32687/2-10 |
| 10941915-11 | M32687/2-11 |
| 10941915-12 | M32687/2-12 |
| 10941915-13 | M32687/2-13 |
| 10941915-14 | M32687/2-14 |

/1 The difference between the parts from the reference and new parts is the finish.
/2 Reference part numbers that are zinc plated with hexavalent chromium are not recommended for new design on Army ground systems.
/3 Use new part numbers in place of reference part numbers with design authority approval.
/4 Zn-HC= Zinc plated with hexavalent chromium
Zn-Ni = Zinc-nickel plated

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 70

Review activities:
Army – AR, AV, CR4, MI
Navy – OS, SH
Air Force – 11, 20

Preparing activity:
Army – AT

(Project 5310-2026-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:34Z
Check the source to verify that this is the current version before use.